WORKED SOLUTION

STEP I 1990, Question 14

Probability and Statistics • Expectation and Optimal Stopping

STEP I 1990, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s1-q14

The key idea

Each draw is independent, because the ball is replaced, so every draw is white, red or black with the same fixed probabilities. For Juggins we only need the chance of surviving three draws and the average score of a surviving draw. For Muggins the question is a trade-off: one more draw adds a little on average but risks losing everything already banked, and the size of that risk grows with N. Comparing "expected score if we draw" with "score if we stop" gives a threshold.

Solution

On any draw let W, R, B be the events that the ball is white, red or black. Since the ball is replaced, every draw is independent of the others, with

$$\mathrm{P}(W) = \tfrac{5}{10} = \tfrac{1}{2}, \qquad \mathrm{P}(R) = \tfrac{3}{10}, \qquad \mathrm{P}(B) = \tfrac{2}{10} = \tfrac{1}{5}.$$

Juggins' strategy

Juggins finishes on zero exactly when a black ball turns up among his first three draws. Otherwise all three draws are non-black and he stops. So

$$\mathrm{P}(\text{final score } 0) = 1 - \mathrm{P}(\text{three non-black draws}) = 1 - \left(\tfrac{4}{5}\right)^3 = 1 - \tfrac{64}{125} = \tfrac{61}{125}.$$

For the expected score, let S be his final score. If the first three draws are all non-black (probability $\frac{64}{125}$), then $S = X_1 + X_2 + X_3$, where X_i is the score of the ith draw given that it is not black. Given that a draw is not black, it is white with probability $\frac{1/2}{4/5} = \frac{5}{8}$ and red with probability $\frac{3/10}{4/5} = \frac{3}{8}$. Hence

$$\mathrm{E}(X_i) = 1 \cdot \tfrac{5}{8} + 2 \cdot \tfrac{3}{8} = \tfrac{11}{8}.$$

In every other case $S = 0$. Therefore

$$\mathrm{E}(S) = \tfrac{64}{125} \cdot 3 \cdot \tfrac{11}{8} = \tfrac{264}{125} = 2.112.$$

Ask yourself Why do we use $\frac{5}{8}$ and $\frac{3}{8}$ here, rather than $\frac{1}{2}$ and $\frac{3}{10}$?

We have already conditioned on surviving, so each of the three draws is known to be non-black, and the probabilities must be rescaled to add to 1. Equivalently, without conditioning: $\mathrm{E}(S)$ is the sum, over the $2^3 = 8$ white/red sequences, of (probability)×(score). Each draw contributes $1 \cdot \frac{1}{2} + 2 \cdot \frac{3}{10} = \frac{11}{10}$ in expectation, and it is kept only if the other two draws also survive (probability $\frac{16}{25}$). This gives $3 \cdot \frac{11}{10} \cdot \frac{16}{25} = \frac{264}{125}$ again.

Muggins' decision

Suppose Muggins has N points and draws once more (and then stops). His score becomes $N+1$, $N+2$ or 0, so

$$\mathrm{E}(\text{score after one more draw}) = \tfrac{1}{2}(N+1) + \tfrac{3}{10}(N+2) + \tfrac{1}{5} \cdot 0 = \tfrac{4}{5}N + \tfrac{11}{10}.$$

Drawing beats stopping on average exactly when

$$\tfrac{4}{5}N + \tfrac{11}{10} > N \iff \tfrac{11}{10} > \tfrac{1}{5}N \iff N < 5.5.$$

The average gain from a draw is $\frac{11}{10}$, and the average loss is $\frac{1}{5}N$, the black-ball chance times what is at stake. Since scores are whole numbers, this suggests the strategy:

keep drawing while the score is 5 *or less; stop as soon as it reaches* 6 *or more.*

Ask yourself Comparing just one more draw with stopping looks short-sighted. Could it pay to draw at $N \geq 6$ and plan to draw again later?

No. Once the score is 6 or more, any further draw leaves it at 6 or more (or ends the game at 0), so at every later decision the one-draw comparison again says "stop". Each extra draw from such a score lowers the expected final score, so the best plan from there is to stop at once. Below 6, one more draw already beats stopping, and the option of carrying on can only add to that. So the threshold rule is the best possible strategy, not just a sensible one.

Using this strategy, let $V(N)$ be the expected final score starting from N. Then $V(N) = N$ for $N \geq 6$, and $V(N) = \frac{1}{2}V(N+1) + \frac{3}{10}V(N+2)$ for $N \leq 5$. Working downwards:

N	6	5	4	3	2	1	0
$V(N)$	6	5.1	4.35	3.705	3.1575	2.690	2.292

So Muggins averages about 2.29 points a game, compared with Juggins' 2.112.

Checks

- A computer enumerated every sequence of up to three draws exactly. It confirmed $\mathrm{P}(0) = \frac{61}{125} = 0.488$ and $\mathrm{E}(S) = \frac{264}{125}$. A Monte Carlo run of 2×10^5 games gave 0.488 and 2.111.

- At $N = 5$ a draw gives $\frac{4}{5} \cdot 5 + 1.1 = 5.1 > 5$. At $N = 6$ it gives $5.9 < 6$. So the switch is between 5 and 6.
- Value iteration over all strategies gave the same rule. Stopping at ≥ 5 or at ≥ 7 instead gives a lower average, 2.261 and 2.265, against 2.292 for stopping at 6.

Common slips

- Calculating $\mathrm{E}(S)$ as $3 \times \frac{11}{10}$, which forgets that a later black ball wipes out the earlier points.
- Multiplying the conditional mean $\frac{11}{8}$ by the unconditional probabilities as well, which counts the survival factor twice.
- Writing the threshold as $N \leq 5.5$ without noting that N is an integer, so the rule is "draw at 5, stop at 6".

Beyond STEP

Muggins' problem is an *optimal stopping* problem. The rule "stop when stopping now beats one more step" is the *one-step look-ahead* rule, and it is optimal whenever, as here, the set of states where it says stop can never be left. The same ideas lead to Markov decision processes and dynamic programming. Grimmett and Welsh, *Probability: An Introduction*, covers the expectation and Markov chain background, and Blitzstein and Hwang, *Introduction to Probability*, has many similar first-step analyses.

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